

Support Vector Machines

DSC 340 Applied Machine Learning

March 17

Linear classification models revisited

- **Logistic regression**

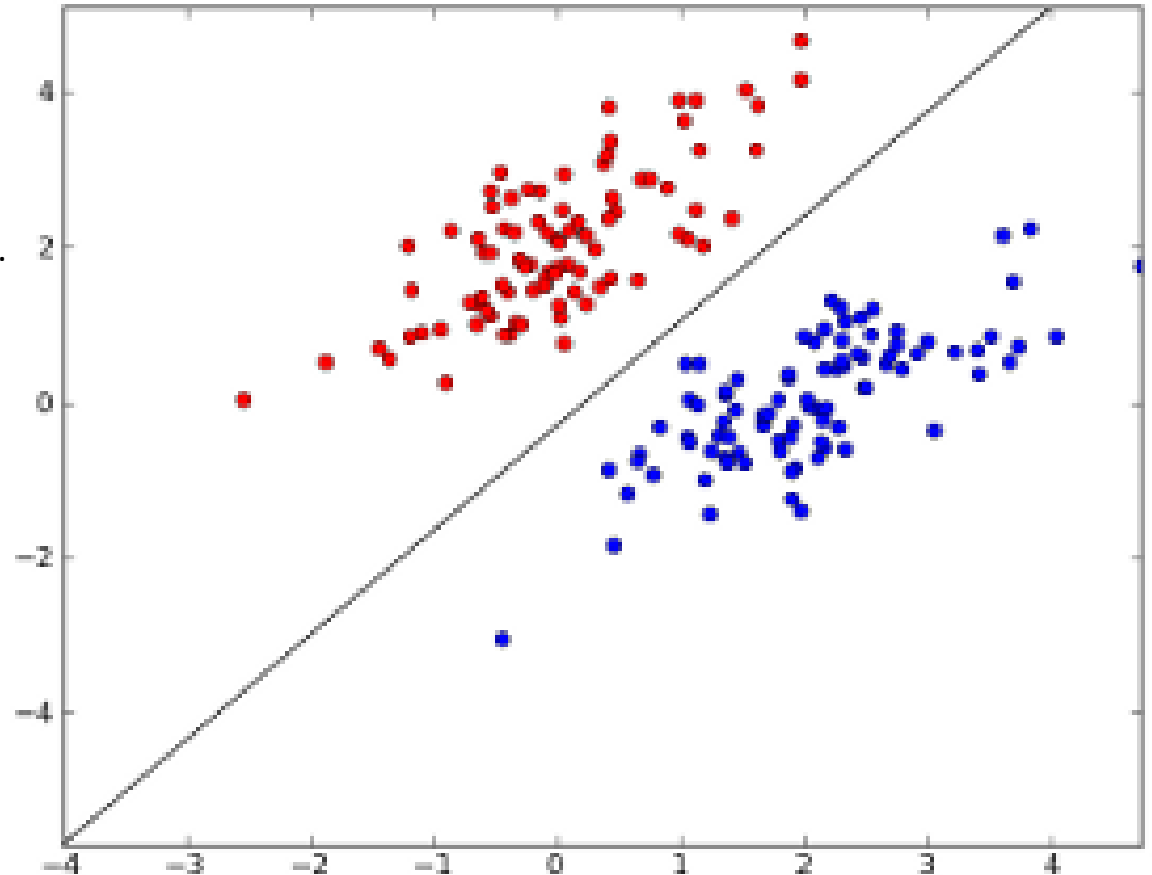
$$P(1|\mathbf{x}) = \frac{1}{(1 + e^{-z})}, \quad z = \mathbf{w} \cdot \mathbf{x} + w_0$$

Note: Linear regression minimizes the **mean squared error** (MSE).
Logistic regression finds a boundary that maximizes the **probability (likelihood)** of correctly classifying observed data.
Equivalently, it minimizes the **log-loss** (also called **cross-entropy**).

- Soon: Perceptron

- Decision boundary: **hyperplanes**

- 1 feature: point
- 2 features: line
- 3 features: plane
- m features: $m - 1$ dim hyperplane



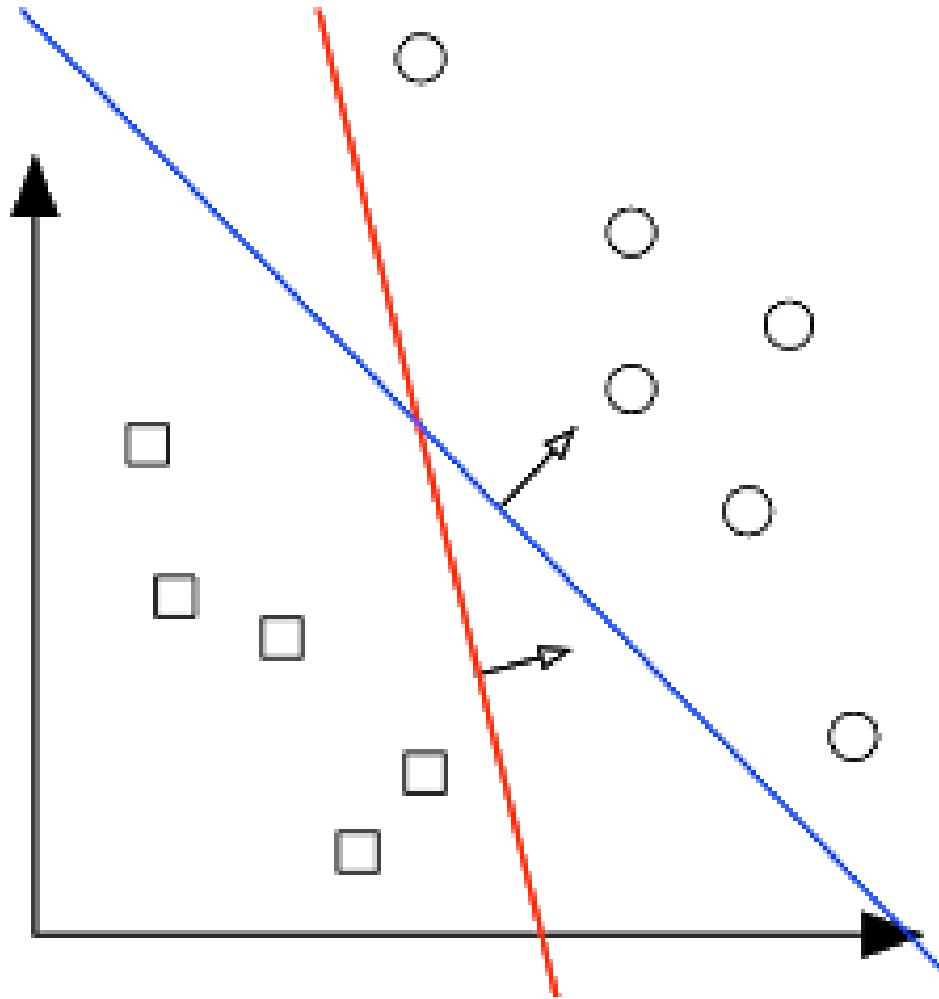
Not linear models

- KNN classifiers
- Decision trees
 - Splits often linear, but recursively applied
- Random forests
- Soon: Naïve Bayes classifiers (NLP/text classification)

SVM Big Idea

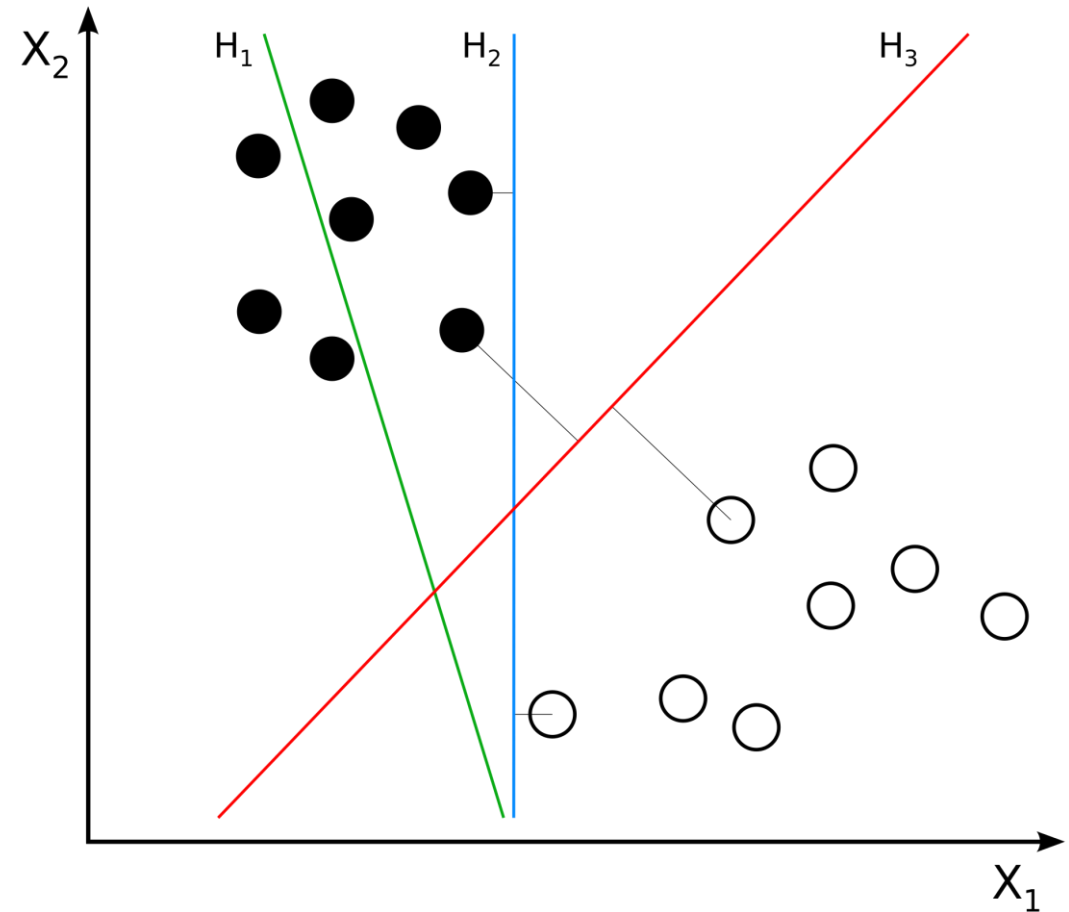
- Transform features
 - New coordinate system with more dimensions
 - For now, think of adding columns that are functions of existing columns
- Maximum margin classifier

Maximum margin classifier

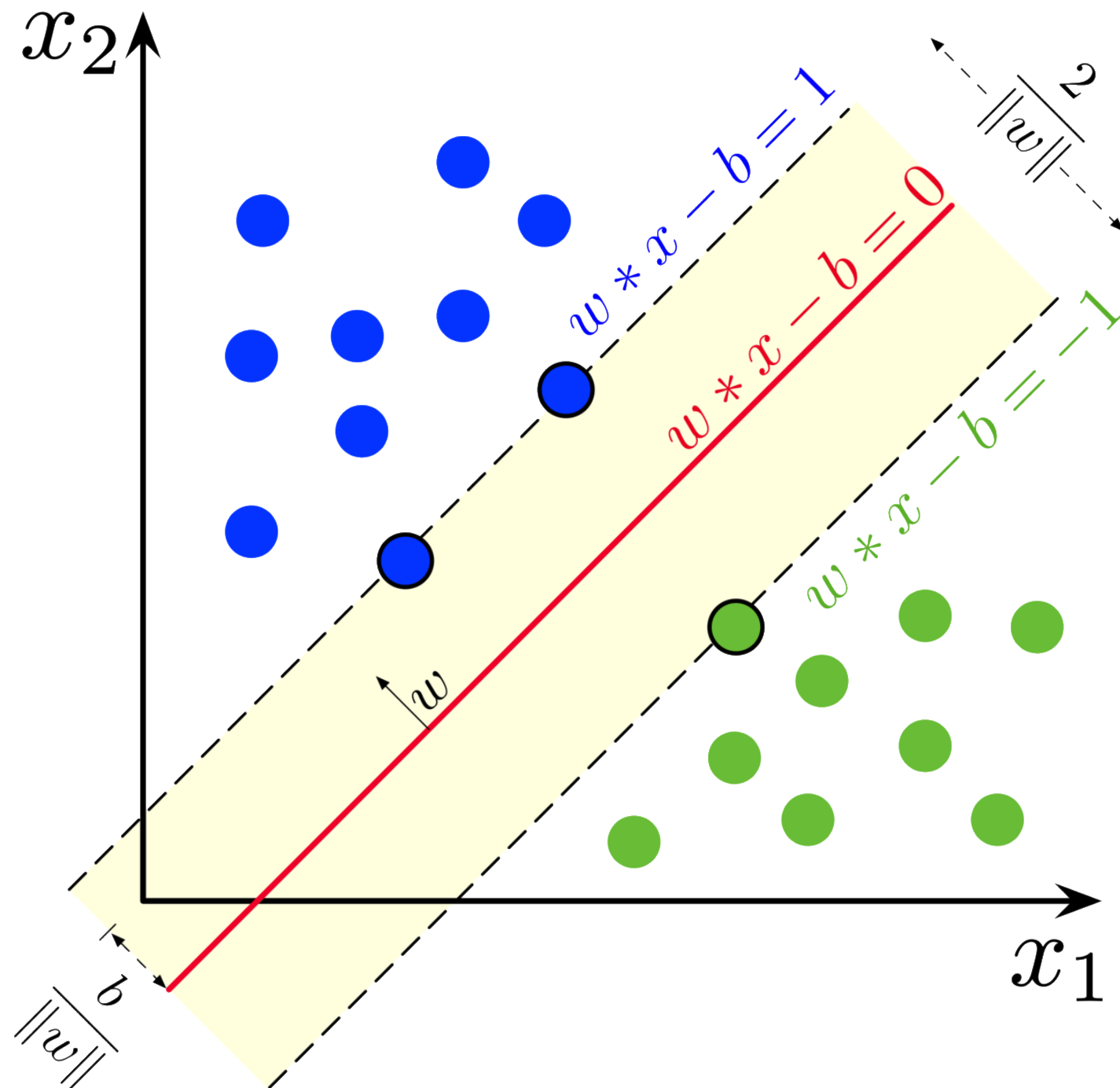


Maximum margin classifier

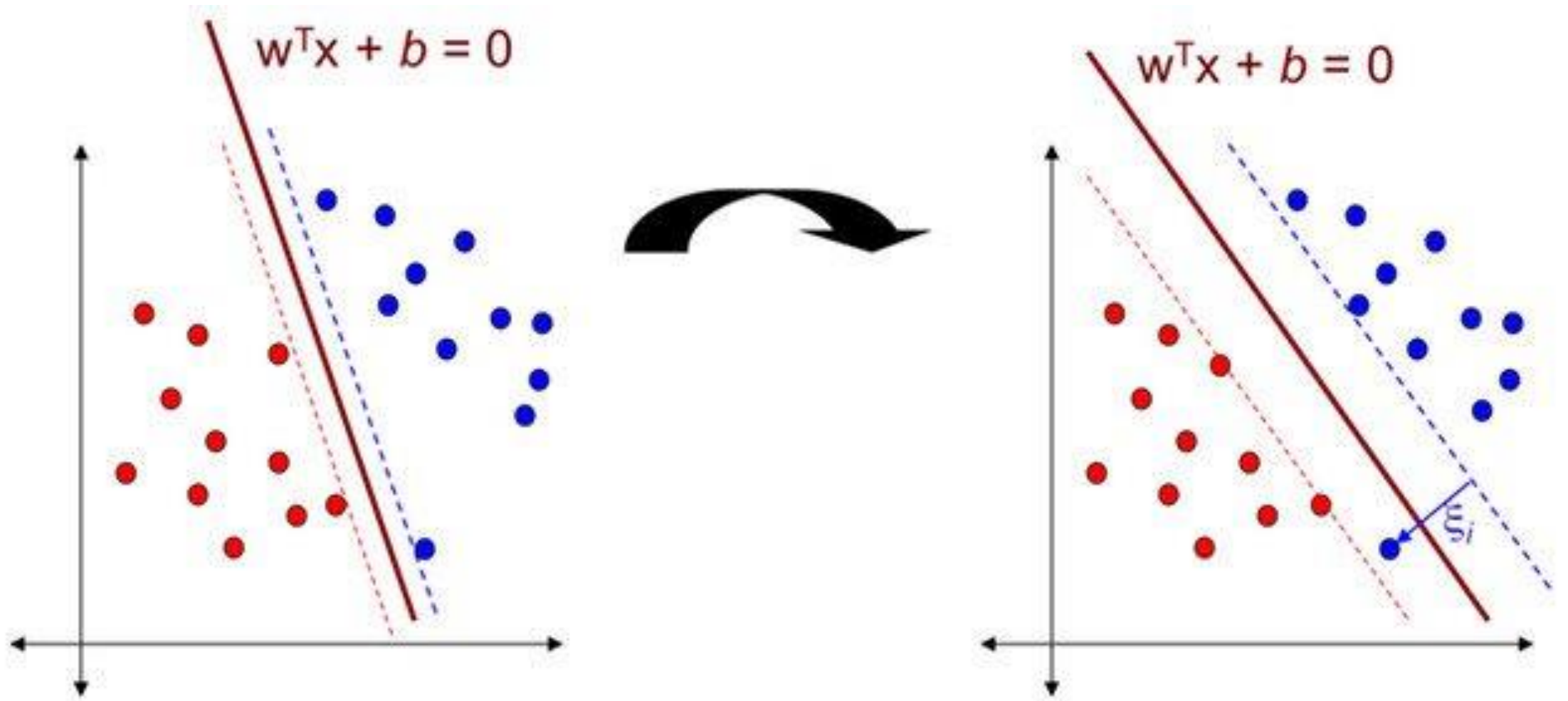
- **Margin ...**
*Distance of **point** ...*
to decision boundary
- Maximum margin classifier
- Reasonable choice of h-plane
- Note: In general, different from:
 - logistic regression
 - perceptron learning algorithm



Support vector classifier



Soft margin



Soft margin

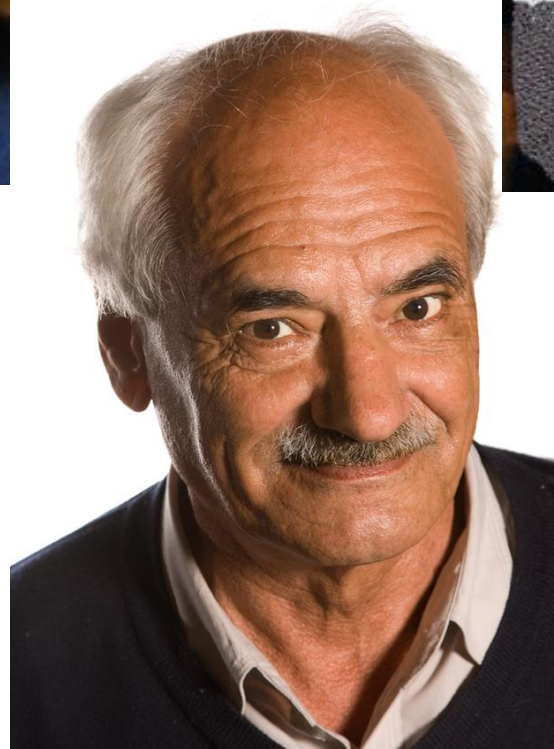
- Reason: **maximum margin classifiers** susceptible to **outliers**
- **Soft margin** allows **misclassifications** (intentionally!)
- Necessary if data not **linearly separable**
- **Bias-variance tradeoff**
- Can explicitly **balance** these for a particular problem: **hyperparameter**

History

- Vladimir Vapnik
- Alexey Chervonenkis
- Isabelle Guyon:

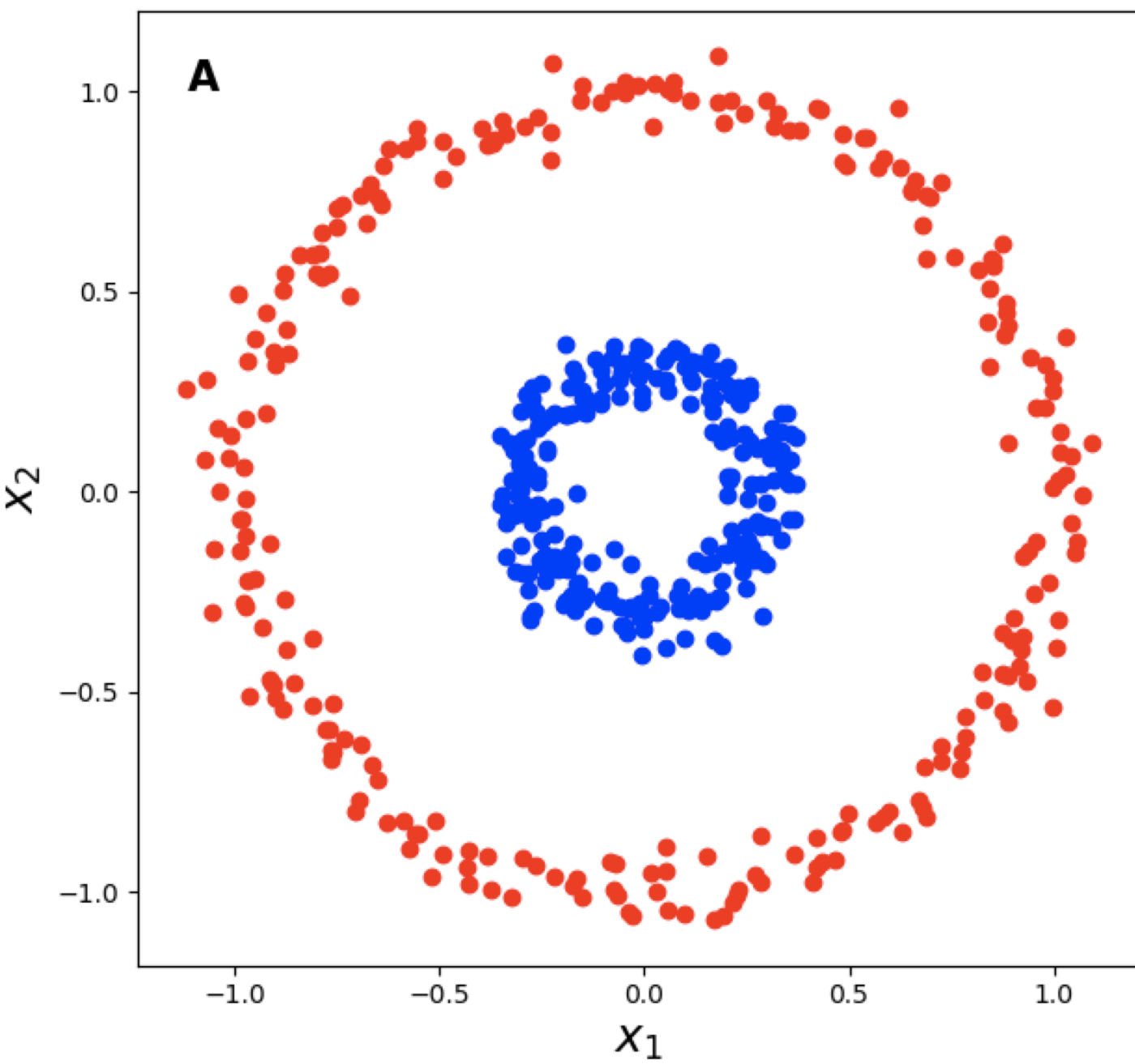


- Idea: use a **kernel trick**!









Kernel trick

- **Lift data** (project) into high (or infinite!) dimensional space
- Contrast: project onto **subspace**
- **Basis functions**
- Lifting: computational expensive
- Certain functions let us work in higher dimensional space ...
- without actually computing ...
- the transformed feature values in the space.
- Uses a **dot product: kernel trick**.

Mathematical Basis

- **Dot product** of vectors $\mathbf{a} \cdot \mathbf{b}$

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots + a_M b_M = \sum_{j=1}^M a_j b_j = \mathbf{a}^T \mathbf{b}$$

- Important to all linear models
- **Kernel function** k can be any function that satisfies the property:

$$k(\mathbf{x}, \mathbf{x}') = \varphi(\mathbf{x}) \cdot \varphi(\mathbf{x}')$$

- *Why is this useful?*
- *Kernels give a way to compute dot products in some feature space ...*
- *without even knowing what this space is and what is $\varphi(\mathbf{x})$.*

Why kernel tricks are useful

- By lifting data into high dimensional space ...
- Better linear separability
- Computationally efficient due to kernel trick
- Doesn't require manually trying a bazillion basis functions
- Does require selecting the type of kernel
- More complex model: still parametric, but a LOT of parameters

Famous kernels you should know

- linear
- polynomial
- radial basis function: RBF
- there are many more!

$$k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}')$$

Not on test. ☺

$$k(\mathbf{x}, \mathbf{x}') = (\mathbf{x} \cdot \mathbf{x}')^d$$

$$k(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2), \gamma > 0$$

- "How to Select SVM kernels" (links on Moodle)
 - <https://www.kdnuggets.com/2016/06/select-support-vector-machine-kernels.html>

Q.: Are SVMs linear models?

- A.: Yes and no.
- If linear kernel, yes.
- If we mean, can they only classify linearly separable data, no.
- Can find decision boundaries in the original space that are not linear.
- But linear in the transformed feature space.

Parameters and hyperparameters

- Parameters: support vectors (in the transformed space)
- Hyperparameters:
- Choice of kernel
- Regularization constant C representing bias-variance tradeoff
- There are many more ...
- See resources posted to Moodle

Assessment

Advantages

- Can learn non-linear boundaries
- Wide selection of kernels
- Robust against overfitting
- Even in high-dimensional space
- Even when columns > rows
- Preferred ML method in 1990s

Disadvantages

- Memory intensive
- Finding right kernel can be tricky
- Don't scale well to large data sets
- Other methods preferred today
 - Random forests, neural nets